



Course MIS 6309.0W1
Course Title Business Data Warehousing
Professor Kashif Saeed
Term Fall 2018
Meetings Online

Professor's Contact Information

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Office Hours Via emails only
TA Information TBD
TA Office hours: Will be posted on eLearning

General Course Information

Pre-requisites, Co-requisites, & other restrictions None

Course Description The course covers traditional (non-SAP) data warehousing concepts. We cover Data Modeling (ER and Ralph Kimball Dimensional Modeling) and Business Intelligence in this class. ETL is not covered in this course.

SLO1: Students will be able to describe architecture and methods for storage and provision of enterprise data.

SLO2: Students will develop competency in query development and essential business intelligence reporting.

Learning Outcomes **SLO3:** Students will demonstrate competency in data modeling, including dimensional modeling.

SLO4: Students will learn steps involved in development of an enterprise data warehousing solution and at least one BI tool from end-to-end perspective.

Optional Texts & Materials

1. Database Refresher and ER Modeling: Database Concepts by David M. Kroenke & David Auer
2. Dimensional Modeling: Data warehouse Lifecycle Toolkit by Ralph Kimball
3. Business Objects: SAP Business Objects BI4 – The Complete Reference by Cindi Howson

1. MySQL and MySQL Workbench
Download Link will be provided on eLearning

Software Used

2. Microsoft Access
Need to be purchased. University has discounted price.

3. SAP Business Objects BI4.2

Download Link: Will be posted on eLearning

4. Tableau Client (the version provided by Tableau)

Download Link: Will be posted on eLearning

Assignments & Academic Calendar

Week	Lectures	Topics to be covered	Description/Tasks	Assignments
1	Week of 8/20 Unit 1 Part 1 and 2	• Introductions and course details • Syllabus Overview and Expectations • Introduction to Data Warehousing • Database Refresher concepts	Tasks: - Install MySQL and MySQL Workbench * Install Instructions will be shared.	None
2	Week of 8/27 Unit 2 Part 1 and 2	• Database refresher concepts - continued • ER Modeling and database design • Classroom Examples and Hands-on Book: Database Concepts by Kroenke and Auer	The purpose of this class is to refresh database concepts so that those with no database background can catch up with the rest of the class. Students will learn ER modeling and how ER models are translated into physical database design.	Assignment # 1 Creating ER model and implementing it using MySQL. Due Date- 9/12
3	Week of 9/03 Unit 2 Part 3 Unit 3 Part 1 and Part 2	• Case for Dimensional Modeling • ER vs. Dimensional Modeling • Dimension and Fact tables • Bus Architecture • Conformed dimensions • Time Dimension • Null handling • Classroom Hands-on: Design your Time and Conformed Dimensions. Book: Data Warehouse Lifecycle Toolkit by Ralph Kimball	Starting from 3 rd class we start Dimensional Modeling using Ralph Kimball methodologies. We will spend roughly 3 classes on Dimensional Modeling.	
4	Week of 9/10 Unit 4 Part 1 and Part 2	• Slowly Changing Dimensions - Type 1, Type 2, and Type 3 • Classroom Hands-on – Design a Type 2 SCD • Role Playing Dimensions • Fact tables • 4 Step Design process • Classroom Hands-on – Design your first Fact table • Classroom Hands-on 2 – Design a multi-star schema Dimensional Model.		
5	Week of 9/17	• Fact less facts		Assignment#2

	Unit 4 Part 3,4,5	• HR Data warehouse review • Snapshot facts • Classroom Hands-on – Design a fact less fact table		Creating a Dimensional Model Due date:10/08
	9/21-9/25	• Mid-term Exam 1 Testing Center 9/20 - 9/22	Students need to register with Testing Center. The exam date can change based on testing center availability	
6	Week of 10/1 Unit 5	BI - The Big Picture • BI Market Analysis • Business Objects Philosophy • Business Objects tool suite • Business Objects timeline • Deployment overview • Job roles in Business Objects • Career options for me in BI • Latest trend in BI job market	First class on Business Objects will cover the big picture explaining the tool suite, when to use which tool, the job roles, deployment scenarios, and the architecture of Business Objects. We will also talk about the career options and latest trends in BI job market.	
8	Week of 10/8 Unit 6 Part 1, Part 2, and Hands-on	• What is a Universe • A well designed universe • Universe Development Process • Local vs. Repository Universes • Getting familiar with the Universe design tools • Joins • Defining Objects and Classes • Classroom Hands-on – Creating my first Universe	All Universe design work will utilize MS Access databases. However, the instructor will give you plenty of information on steps involved with Oracle and SQL Server based universes.	
9	Week of 10/15 Unit 7 Part 1, Part 2, and Hands-on	• What are Loops • OLTP vs. Star schema based universes • Resolving Loops • Classroom Hands-on - Create a Dimensional Model schema based Universe and Resolve loops • Classroom Hands-on – Create an ER Model schema based Universe and Resolve loops	This class will cover the most important topic in Universe design – loops and how to resolve them.	
10	10/19-10/23	Midterm Exam 2 Location: Testing Center 10/18-10/20	Students need to register with Testing Center. The exam date can change based on testing center availability	
10	Week of 10/22 Unit 7 Part 3,4, and Hands-on	• Identifying Chasm traps • Resolving Chasm traps • Identifying Fan traps • Resolving Fan traps • Classroom Hands-on - Resolve traps	This class will cover Traps and how to resolve them. We will do hands-on exercises in the class.	Assignment#3 Universe Design Due Date: 11/07
11	Week of	• Hierarchies		

	10/29 Unit 8 Part 1 & 2	• Universe level restrictions • Row level security • Using IDT (Information Design Tool) • Classroom Hands-on – Building a Universe with IDT		
12	Week of 11/5 Unit 8 Part 3 and 4	• Web Intelligence usage as reporting tool • Building reports with conditions • Publishing and sharing reports • Sections, breaks, calculations • Report level filters • Classroom Hands-on	The purpose of this class is to teach fundamentals of reporting using Web Intelligence. We will use examples to cover the concept.	
13	Week of 11/12 Unit 9 Part 1,2,3	• Webi Report variables • Webi Multiple data providers • Classroom Hands-on – variables and Multiple DP		Assignment # 4 – building report using Web Intelligence and Tableau Due Date: 12/04
14	Week of 11/19	Fall/Thanks Giving Break	No Class	
15	Week of 11/26 Unit 10 Part 1 and Part2	• Tableau background • Getting started • Connecting to data sources • TDS and TDE files • Publishing data sources • Filtering and Quick Filtering Groups • Hierarchies • Sets • Working with Dates • Building dashboards • Interactivity using actions • Sharing your dashboards • Data Blending		
16	11/30-12/4	• Final Exam (Comprehensive) Location: Testing Center 11/29-12/1	Students need to register with Testing Center. The exam date can change based on testing center availability.	

Course Policies

Grading (credit) Criteria	The course uses RANK based grading Top 25% Students – A Next 15% Students– A-
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	Next 20% Students – B+ Next 20% Students - B Remaining Students – Other grades Grading Weight Assignments – 25% Midterm 1 – 25% Midterm 2 – 20% Final – 30%
Testing Center Guidelines	To provide our students a successful testing experience we encourage them to be familiar with the following guidelines: - Students taking exams <u>on campus at UT Dallas Testing Center:</u> - Please direct your students to register for <u>ALL</u> their exams for the entire semester during the first two weeks of semester, or no later than 72 hours prior to exam date (based on seat availability, seats might be available if they wait until the last minute), via this link https://ets.utdallas.edu/testing-center. - We strongly encouraged students to read the “Student Guidelines” prior to taking exams https://ets.utdallas.edu/testing-center/students/ - Walk-ins will not be allowed, <u>students must have an appointment to be admitted</u>. - Students are <u>required</u> to present a <u>physical</u> Comet ID Card to be admitted, no exceptions - Students taking exams off campus at a <u>remote testing center:</u> - All Proctor Exam Application request must be submitted at least 15 business days prior to the exam date or you will need to reschedule for another exam date within the professor approved exam window.
Make-up Exams	There will be no make-up exams. However, I will work with you if you have a date conflict and would like to change the date of your exam.
Extra Credit	None
Late Work	Penalty on late assignments will be listed on the Assignment itself.
Special Assignments	None
Class Attendance	Optional
Classroom Citizenship	- Use of cell phone not allowed during the class and the exams. - Cheating during the exams and/or assignments will not be tolerated. - Standing outside of testing center and sharing the questions that appeared in your exam with other students is considered cheating. If caught, your case will be turned over to the Judicial Affairs department. Class Quizzes will require students to install Lockdown browser. http://www.utdallas.edu/elearning/resources/documents/ldb_student.html
Comet Creed	<i>This creed was voted on by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:</i> <i>“As a Comet, I pledge honesty, integrity, and service in all that I do.”</i>
UT Dallas	<i>The information contained in the following link constitutes the University’s policies</i>

Syllabus Policies and Procedures	<i>and procedures segment of the course syllabus.</i> <i>Please go to http://go.utdallas.edu/syllabus-policies for these policies.</i>
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The descriptions and timelines contained in this syllabus are subject to change at the discretion of the Professor.